

An Intelligent Robot Sorting System By Deep Learning On RGB-D Image

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Abstract—Industrial robots have seen wide adoption in the field of intelligent manufacturing, and they are applied to the intelligent sorting of items on the industrial product line. However, such techniques have some limitations, such as the object being placed at a fixed height. To solve this problem and meet the needs of flexible manufacturing, this paper develops an intelligent robot sorting system by deep learning on RGBD image (RSSDR). RSSDR consists of five modules, including central host, robot, visual kit, microservices and machine vision control software. The central host provides the image of 3D printed industrial parts collected by the visual kit, and calls the deep learning algorithm on the microservice to detect the key points on the image, and then sends the position of the grasping points to the robot through the machine vision control software to control the robot to sort the 3D printed industrial parts. In RSSDR, to visually identify a total of 3 different 3D printed industrial parts in uncontrolled conditions, YOLOv5s is used as the detection model, which achieves 97.65% accuracy and 35ms runtime. The experimental results show that RSSDR is suitable for sorting electronic component.

Keywords—component; intelligent manufacturing; flexible manufacturing; intelligent robot sorting system; microservice; machine vision; YOLOv5s

I. INTRODUCTION

E-commerce logistics has driven the development of many industries across the country. The time from production to packaging to customers is getting shorter and shorter, and the time left for factories is constantly shrinking, and factories urgently need automated sorting equipment. In the development of industrial production, people can often see the shadow of sorting robots in industries such as medicine, food, chemicals, and logistics. In particular, the capacitance is relatively small, and manual sorting and stacking are difficult, which often requires refined instruments. Under high-intensity labor, people are prone to fatigue and stacking errors, which will have a negative impact on subsequent use.

Over the years, a lot of research has been done on sorting robots. The working process of the sorting robot is that under the guidance of machine vision, the robot end effector reaches the corresponding position and grabs the target object to the designated position. First, the vision camera is calibrated to

accurately capture the image of the target object. Secondly, image processing is performed to determine the coordinates and characteristics of the target object through image processing, and finally the information is converted into the coordinates and attitude of the end effector to ensure accurate and efficient completion of the sorting work[1].

This article designed and completed a robot grabbing electronic component system based on RGBD camera. With the help of visual components, the robot has realized the tasks of identification, classification, grabbing, and placing 3D printed industrial parts. Among them, the image of the detection and identification of 3D printed industrial parts was captured by the RGBD camera. And use YOLOv5[2] to identify 3D printed industrial parts based on YOLOv5 to achieve high accuracy. The structure of the rest of the paper is organized as follows: Section II introduces the overall design of the system including the architecture of RSSDR and the details of the five modules; Section III, the overall system approach; Section IV describes the hardware and software components of RSSDR And the performance of the YOLOv5 model and the experiment of RSSDR; Section V, the conclusion is given.

II. OVERALL DESIGN OF RSSDR

The overall structure of RSSDR is shown in Fig. 1(a). RSSDR is mainly composed of five modules, including the central host, robot, visual kit, microservices and machine vision control software. The central host is designed to control every process of the RSSDR and it serves the communication of the robot, camera and machine vision control software. The main task of the robot is to grab the 3D printed industrial parts on the platform, identify, classify, and carry out the grabbing task with the help of the vision component. Vision components need to capture images of 3D printed industrial parts through cameras, detect 3D printed industrial parts on the platform and identify 3D printed industrial parts. Microservices provide a storage platform for algorithms and images. Machine vision control software builds test platforms for robots, microservices and cameras, including displaying images of 3D printed industrial parts and controlling robot motion strategies.

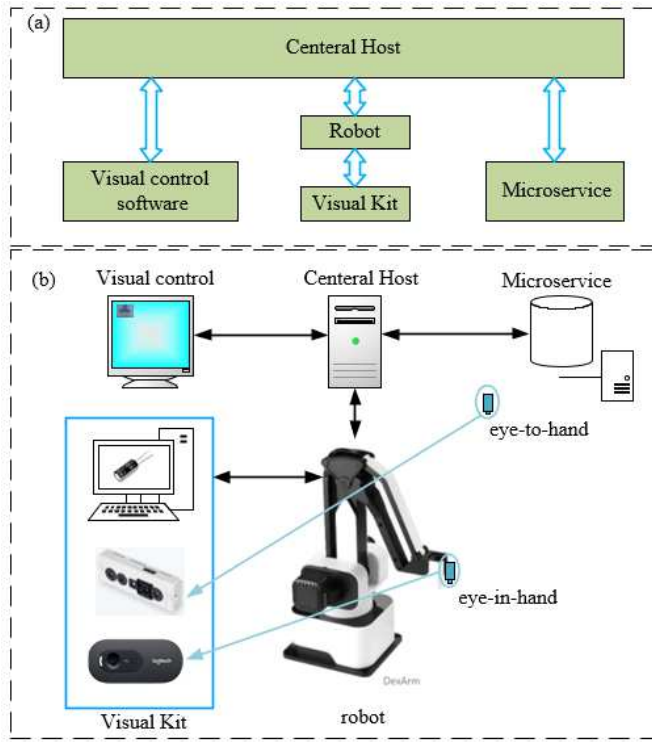


Figure 1. The Overall Architecture of Designed RSSDR.

The detailed design of RSSDR is shown in Fig 1(b). The robot is a visual task with the help of a visual kit composed of a host and two cameras. One camera (camera 1) is installed on the robotic arm for correction detection of 3D printed industrial parts, and the other camera (camera 2) is fixed at 40cm height to identify and classify the 3D printed industrial parts. At the begin, the mechanical arm movement to a fixed position, and the RGBD camera obtains deep images and color images. Next, accurately capture the location information of the electronic component through the YOLOv5 model, convert the deep camera coordinates into a robotic arm coordinates, send it to the host, and at the same time, the robotic arm movement To the top above the target, the monocular camera shooting image is determined to grasp the target position again, and then the host sends the robotic arm instruction to control the robotic arm to achieve the crawling task.

III. METHODS

Mainly by the algorithm function processing flow of RSSDR as shown in Fig 2 and the electronic device detection model based on YOLOv5 as shown in Fig3.

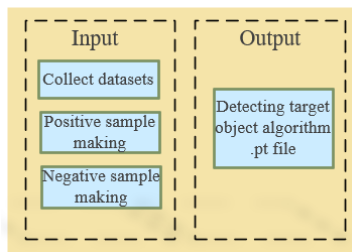


Figure 2. Electronic Device Detection Model Flow Based On YOLOv5

A. The Algorithm Function Processing Flow Of RSSDR

1) Image preprocessing module

The quality of the image directly affects the design and accuracy of the detection and recognition algorithm. Therefore, before image analysis[3] (feature extraction[4], segmentation[5], matching[6], and recognition, etc.), preprocessing is required. The main purpose of image preprocessing is to eliminate irrelevant information in images, restore useful real information, enhance the detectability of relevant information, and simplify data to the greatest extent, thereby improving the reliability of feature extraction, image segmentation, matching and recognition. Image preprocessing algorithms mainly include distortion correction algorithm, depth map and color map registration algorithm and depth map completion algorithm.

2) Distortion correction algorithm

Lens distortion[7] is a general term for the inherent perspective distortion of optical lenses, which can be simply divided into radial distortion and tangential distortion. Radial distortion refers to the fact that the actual image point is farther or closer to the image center than the ideal image point, and tangential distortion refers to the angular difference between the actual image point and the ideal image point relative to the image center. The inverse solution is used for distortion. First, coordinate transformation is performed to convert pixel coordinates into picture coordinates. Then, each coordinate on the undistorted image corresponds to the coordinate on the distorted image by adding distortion. Finally, the corresponding relationship obtained in the previous step is the distorted image. Pixels on give the coordinates on the undistorted image.

a) Monocular vision calibration

Use the camera to be calibrated to take a number of chessboard pictures from different perspectives, and calibrate these pictures with MATLAB toolbox to calculate the internal parameters of the camera and the external parameters corresponding to each image[8].

b) Depth camera calibration

The external infrared light source illuminates the chessboard image, and at the same time, the infrared transmitter of the Gemini camera is completely blocked with black tape. After the depth camera acquires multiple infrared images, the MATLAB toolbox is used to calculate the parameters[9].

3) Depth map and color map registration

The registration of the depth map and the color map In order to fuse the information of the depth map and the color map, the two-dimensional coordinates of the target object are obtained through the color map, and the three-dimensional information of the object can be obtained by obtaining the depth information of the same position from the depth map. The registration method adopts the hardware parameter method. Considering that theoretically, the depth image and the same point on the color image can be obtained only by the positional transformation of translation or rotation. Therefore,

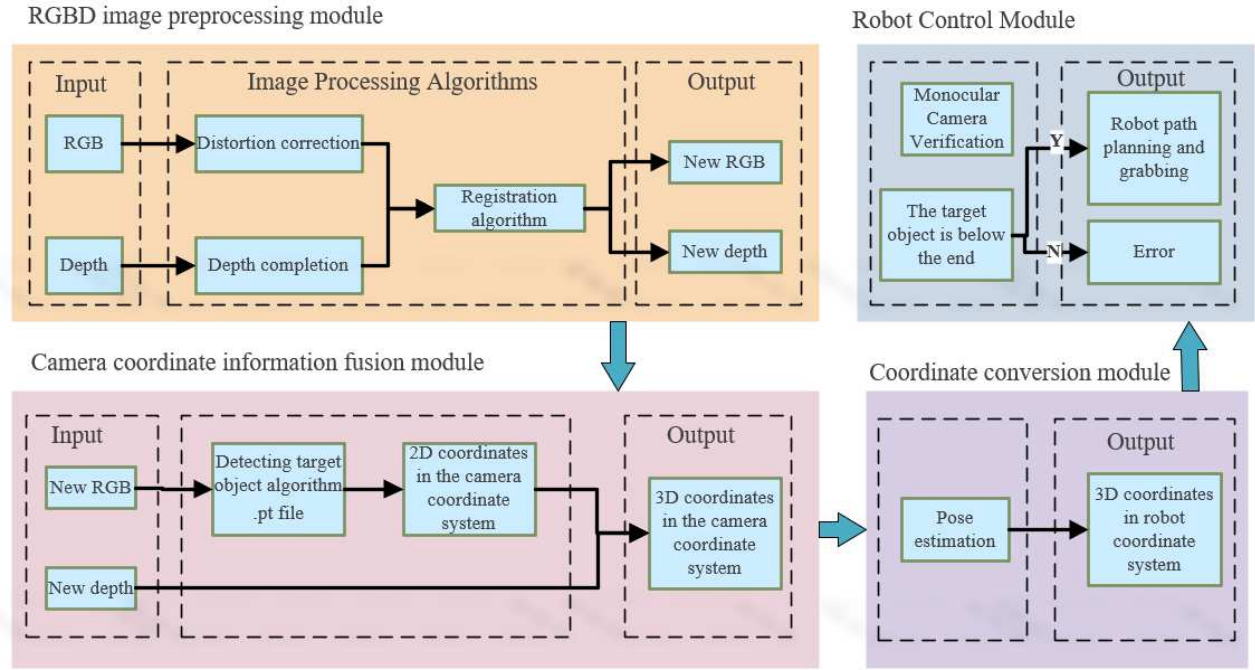


Figure 3. Electronic Device Detection Model Flow Based On YOLOv5

the matrix operation can be used to convert the matrix between the two using hardware. The parameter matrix (internal and external parameters of the camera) is expressed to complete the registration.

$$M_{dc} = A_{dc} \times M_{dp} \quad (1)$$

Equation (1) is depth camera coordinate transformation expression. Suppose M_{dc} is the 3-dimensional space coordinate vector of the point to be measured M in the depth camera coordinate system, M_{dp} is the 3-dimensional coordinate vector of the point in the image coordinate system, A_{dc} is the camera's internal parameter matrix, from the pinhole imaging model.

$$M_{rgb} = M_{rgbp} \times A_{rgb} \quad (2)$$

Equation (2) is RGB camera coordinate transformation expression. Suppose M_{rgb} is the spatial coordinate vector of the same point in the RGB camera coordinate system, M_{rgbp} is the projected coordinate of the point on the RGB image plane, and A_{rgb} is the internal parameter matrix of the RGB camera.

$$M_{dp} = R \times M_{rgbp} + T \quad (3)$$

Equation (3) is transformation formula of depth image and color image. Since the coordinates of the depth camera are different from the world coordinates of the RGB camera, it is abstracted as two points for research, so it can be connected by a rotation and a translation. In the formula: R and T are the registration transformation parameters.

$$\begin{cases} M_{dp} = B_d \times M_w = \begin{pmatrix} R_d & T_d \\ 0 & 1 \end{pmatrix} \times M_w \\ M_{rgb} = B_{rgb} \times M_w = \begin{pmatrix} R_{rgb} & T_{rgb} \\ 0 & 1 \end{pmatrix} \times M_w \end{cases} \quad (4)$$

Equation (4) is the extrinsic parameter matrices B_d and B_{rgb} of the camera, M_w can be converted into camera coordinates M_{dp} .

$$\begin{cases} R = R_{rgb} \times R_d^{-1} \\ T = T_{rgb} - R_{rgb} \times R_d^{-1} \times T_d \end{cases} \quad (5)$$

Equation (5) is the relationship between registration transformation parameters R , T and camera internal and external parameter matrix.

4) Depth completion algorithm

The edge of the RGB image will play a role as a constraint in the experiment. The depth image restoration is performed within the range delimited by the edge of the RGB image, which is equivalent to reconstructing the depth image once, so that it can complete a more accurate position without losing as little information as possible. Transform, align with RGB image.

The depth image part is divided into two steps: coarse restoration and fine restoration. The main method of the former step is filtering, and the latter step is realized by the region growing algorithm. Filtering algorithms are often used

in image restoration with insufficient accuracy, especially in the case of irregular edges that need to be processed this time. The accuracy of the results of the region growing algorithm can be satisfactory, but it is correspondingly lost in the processing speed, so thick and thin, fast and slow, can learn from each other and meet our needs[10].

a) Coarse recovery

Transfer the edge of the color image to the depth map, and then according to the coordinates of the edge of the RGB image, the depth of the depth image near it (in this experiment, the edge point of the color image itself and all the pixels in the four-connected area) values are set to zero. With the pixel as the center, a window with a size of 5×5 is established, all pixels with non-zero gray value in the window are traversed, and the number of occurrences of each gray value is recorded using a hash table.

$$I_{new}(x, y) = \frac{\sum_{i=1}^{255} H_i \times i}{\sum_{i=1}^{255} H_i} \quad (6)$$

Equation (6) is Pixel average expression. Set two thresholds, threshold1 and threshold2, to determine if the number of non-zero pixels on the border in the window is greater than threshold1 and whether the non-zero pixel value on the non-border is greater than threshold2, then calculate the weighted average of the pixels in the window.

b) Fine recovery

The region growing algorithm is an iterative recursive process. The process is as follows: after determining the starting point (called the seed point), the similarity (called the growth criterion), and the growth stop condition, make the image near the seed point, and the pixel points that meet the growth criterion are merged with the seed point to become region, and the boundary point of the region is used as a new growth point to repeat the process until the growth stop condition is reached, which is widely used in image segmentation and other fields. Take any pixel with a depth value of zero (that is, the center point of the skipped filter window in the above rough restoration process) as the seed point. The growth criterion requires that the depth value of the new growth point is also zero and the closest to the previous growth point, the growth stop condition is set as the seed point as the starting point and no new growth point can be found.

5) Result

For the registration results of the above methods in practice Fig 4, Fig 5, Fig 6:



Figure 4. Left is color, right is depth histogram

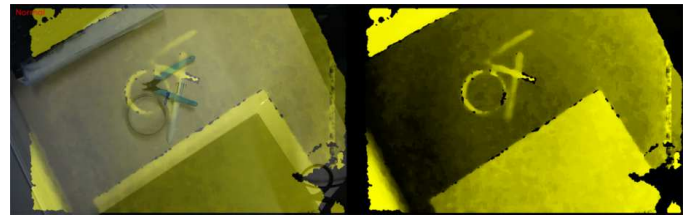


Figure 5. Left is the fusion of the color map and the depth histogram, Right is depth histogram

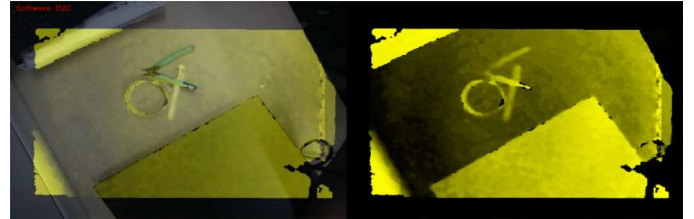


Figure 6. Left is the fusion of the color map and the depth histogram(registration), Right is depth histogram

Fig 4 shows the colour map and depth histogram to be aligned, after fusion as in the left panel of Figure 5, there is a large error between the two, after the above alignment to obtain Fig 6, the colour map and depth histogram are accurately aligned.

B. Image data acquisition and processing



Figure 7. Image Acquisition Hardware-Gemini Camera

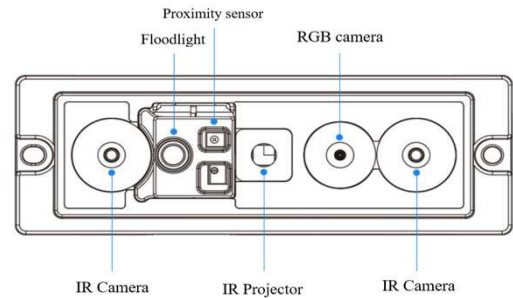


Figure 8. Image Acquisition Hardware Structure Diagram

The image acquisition hardware used in this paper is the Gemini 3D camera shown in Fig 7, which is a close range, high precision embedded module based on binocular structured light 3D imaging technology, capable of outputting higher resolution and higher frame rate depth information, with stable performance and excellent performance both indoors and outdoors, consisting of floodlight, proximity sensor, RGB camera, two IR cameras and IR Projector, as shown in Fig 8.

The YOLO series of algorithms regress the target coordinates and categories end-to-end. The subsequent

improved YOLOv2[11], YOLOv3[12], YOLOv4[13] and YOLOv5 based on YOLO are more optimized in terms of detection speed and accuracy. Considering that the image is mainly from a fixed-height RGBD camera, and the 3D printed industrial parts account for a small proportion of pixels in the image, this paper needs to focus on the detection ability of the detector for small targets when dealing with the problem of target object detection in the image. Based on the performance of classical methods in detection speed and detection accuracy, YOLOv5s is selected as the basic network model for the image target detection task in this paper to improve the detection ability of 3D printed industrial parts in images.

1) Data set

The classification of 3D printing industrial parts mainly includes Component 1, Component 2, and Component 3. As shown in Fig 9. This paper mainly identifies the above three types, and needs to collect a large number of images to create a data set for training. In order to collect enough and balanced images, some data augmentation techniques are used to enlarge the dataset, and the samples of each class are made with roughly the same size. At the same time, the pictures of the three parts under different light conditions were collected, as shown in Fig 10 (a), (b), (c), and (d) respectively. Before training the YOLOv5 model, the model needs to be classified and labeled. The open source image labeling software Labellmg is used in this article. The label type is *Component 1(C1)*, *Component 2(C2)*, *Component 3(C3)*.

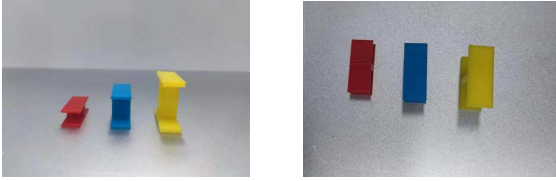


Figure 9. 3D printed industrial parts

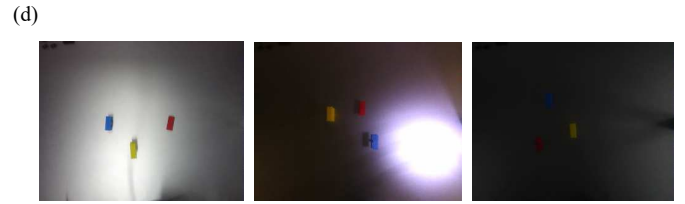
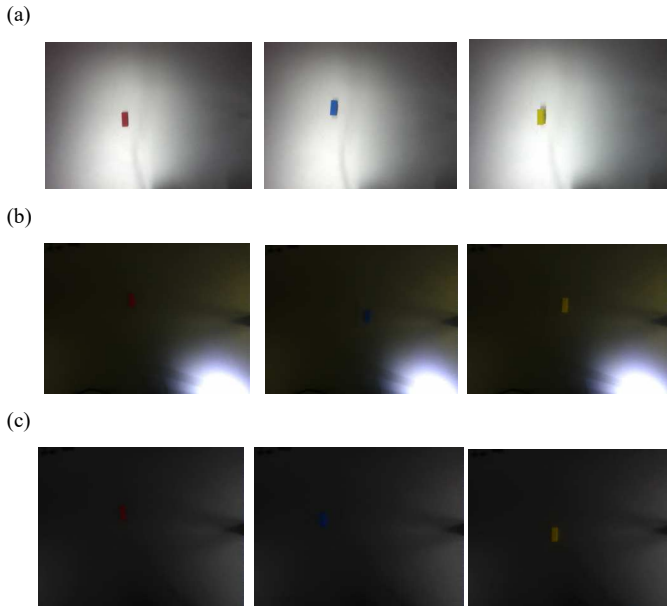


Figure 10. Datasets in Different Environments

2) Electronic device detection model based on YOLOv5

YOLOv5 is divided into four different models: YOLOv5s, YOLOv5m, YOLOv5l and YOLOv5x, among which YOLOv5s has the fastest detection speed and the smallest model file. Since it is to be used in the grasping experiment of the robot, the detection speed and the size of the model should also be considered while considering the accuracy. Therefore, this study uses YOLOv5s, which is easy to embed the system, as the backbone network model[14].

The YOLOv5 model network structure consists of four major parts, as shown in Fig 11, Input, Backbone, Neck and Output. Input adopts the Mosaic data enhancement method and can adaptively scale the picture, and automatically calculate the best anchor frame value of the data set. Backbone is mainly composed of Focus structure and CSPNet structure. The main job of the Focus structure is to slice the image. After four slice operations, a convolution operation with a convolution kernel of 32 is performed, and the original image with a resolution of $640 \times 640 \times 3$ becomes a feature image of $320 \times 320 \times 32$. CSPNet is a cross-stage local fusion network. Its main function is to extract the features of the feature map and obtain the features of different layers to enrich the image information. In order to increase the receptive field and fuse features of different scales, YOLOv5 puts SPP in the Backbone part, which can extract important features. In the Neck part, the structure of FPN+PAN is used, and the FPN adopts the up-sampling method to obtain the spliced feature map, and then aggregates the shallow features from the bottom to the top through the PAN network structure to fully integrate the image features of each layer. The Neck part contains a CSP2 structure. Unlike the CSPNet structure in Backbone, CSPNet contains a residual module, while CSP2 contains a basic convolutional CBS module. In the Output part of YOLOv5, CIOU_loss is used as the loss function, and non-maximum suppression is used to filter out redundant target frames, predict image features and find the best detection position.

C. Coordinate conversion

The experiments were carried out on a completed RSSDR (Fig 12), where the camera provided 3D information about the target object, which was sent to the robot via coordinate transformation to perform a pick-and-place grasp. During the calibration of the camera, we set the world coordinate system to coincide with the robot base coordinate system, as shown in Fig 13 The relative relationship between the pixel and image coordinate systems can be seen.

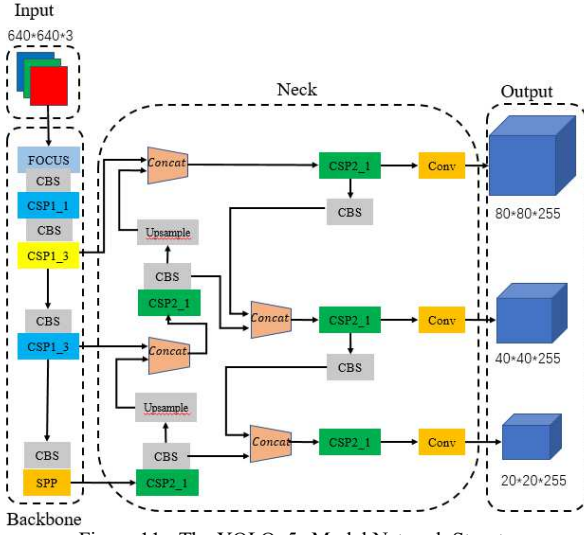


Figure 11. The YOLOv5s Model Network Structure

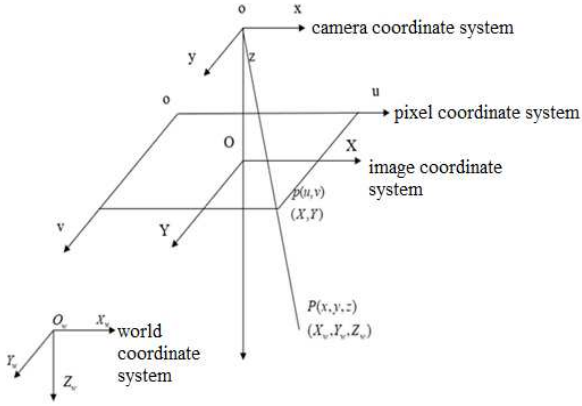


Figure 12. Pinhole Imaging Model

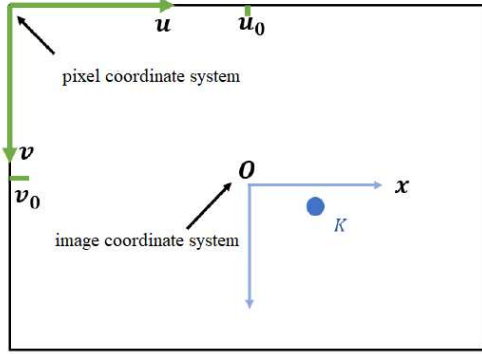


Figure 13. Pixel coordinate system and image coordinate system

Equation (7) can describe the conversion relationship between the world coordinate system and the pixel coordinate system:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} 1/dX & 0 & u_0 \\ 0 & 1/dY & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \quad (7)$$

$$= \begin{bmatrix} \alpha_x & 0 & u_0 & 0 \\ 0 & \alpha_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$= M_1 M_2 X_w$$

$$= M X_w$$

In the formula: α_x —the scale factor of the axis, $\alpha_x = f/dX$; α_y —the scale factor of the axis, $\alpha_y = f/dY$; M_1 —the internal parameter matrix of the camera, referred to as the internal parameter matrix; M_2 —the external parameter matrix of the camera, referred to as the external parameter matrix; M —the projection matrix of the camera.

D. Action decision

A behavioral decision-making strategy based on the Belief-Desire-Intention (BDI) model was adopted in RSSDR. In essence, the problem to be solved is to determine the main goal and how to achieve this goal. The first step is to determine the goal, according to the status quo of the subject's beliefs, select a set of desires that need to be realized. The second step is responsible for determining how to achieve these specific goals resulting from the previous step through the available options.

In the robot grasping experiment, the memory set, the rule set, the behavior set and the state set constitute the robot's belief. Memory sets store events that occurred, including camera detection status. The rule set stores the rules of robot crawling, records the classification rules and the crawling order. Behavior sets are the actions of the robot, such as grasping, carrying, and placing. The state set contains the knowledge of the robot, the sequence of grasping 3D printed industrial parts. The actions of the robot player are determined by the following steps. First, the robot's beliefs are updated based on environmental information. The goal setting is then formed by selecting the state with the least amount of drops and paint textures. Then, decompose the goal to obtain a state that can be achieved in one step. Finally, these states are matched to actions to determine the current action.

IV. RESULTS AND DISCUSSION

Hardware and software are completed in light of the designed RSSDR. The hardware is composed of robot, RGBD camera, computer, ordinary camera and workbench. A robot is a modified robot that grabs and releases 3D printed industrial parts. RGBD cameras are used to capture images in order to classify, identify and locate 3D printed industrial parts. Computers are used to process information and send instructions to robots. Ordinary cameras are used to correct erroneous information and prevent incorrect instructions. The workbench is used to install robots and cameras, and to handle

electronic component platforms. The software part includes client software, which is mainly responsible for communication, control and display of intermediate image results.

A. Hardware

In order to make the robot grasp more stable, the robot has removed the suction cup type on the basis of appearance and structure, and added a flexible gripper (Fig 14(b)) on the basis of the prototype (Fig 14(a)), which can protect the gripping object, with high precision (0.05mm). it needs to 3d print a fixture (Fig 14(d)) to make it fixed on the end of the robot.

As for cameras, RSSDR requires 2 cameras. Considering the installation of the camera, with the help of a 3D printer, a housing (Fig 14(c)) was designed. The camera circuit board is embedded in it, and two rows of LED lights are arranged on both sides of the camera. One camera and housing are at the end of the robot, and the other is at the top of the robot, The schematic diagram of the modified arm is shown in Fig 14(e)).

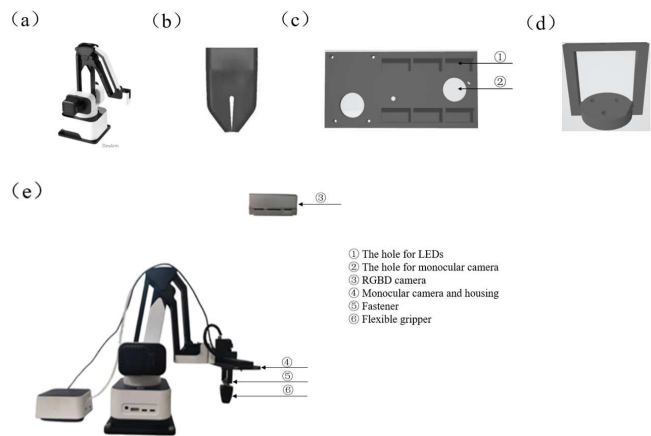


Figure 14. The Main Hardware of Robot. (a) The Prototype of Robot; (b) The flexible gripper; (c) The Model of Designed Housing for Camera; (d) The fixture;(e) The Modified Robot.

B. Software

The main components of the software are the communication module, the interface between the microservice and the machine vision software, the communication module is used for information transfer between the robot, the camera and the machine vision software, the machine vision software and the algorithm in the microservice are connected, the algorithm processing result is returned, and the command is sent to control the robot. The schematic diagram of the software is shown in Fig 15.



Figure 15. The Schematic Diagram of Software in RSSDR

C. Performance of the YOLOv5 model

The performance of the proposed model consists of the identification of 3D printed industrial parts, the accuracy and efficiency are evaluated, and comparative experiments with traditional RCNN are carried out. The test results show that the accuracy of the YOLOv5 model reaches 97.65%, while the RCNN is only 92.32%.

Moreover, the system with the YOLOv5s model deployed is tested, which aims to evaluate the performance of the YOLOv5s model in a relatively real environment. In the test, the 3D printed industrial parts are placed on the workbench, and then detected by the RGBD camera, picked up by the robot, and sorted. Category 3 and negative sample 3D printed industrial parts are tested 300 times, which means a total of 600 times. Table I shows the confusion matrix showing the details of this detector. The results showed that 3 classes were identified with 100% true positive rate, and less than 5 samples were confused in *Component 1(C1)*, *Component 2(C2)* and *Component 3(C3)*,

TABLE I YOLOv5s Detection Results

Class	Objects	Number of false positives	Precision/%
Component1(C1)	201	6	97.0
Component2(C2)	199	5	97.5
Component3(C3)	197	3	98.5

V. EXPERIMENTS AND CONCLUSIONS

The experiments were carried out on a completed RSSDR (Fig 16), where the camera provides 3D information about the target object, which is sent to the robot via coordinate transformation to perform a pick-and-place grasp. a series actions including detecting the accessible component (Fig 17), grasping it (Fig 18).

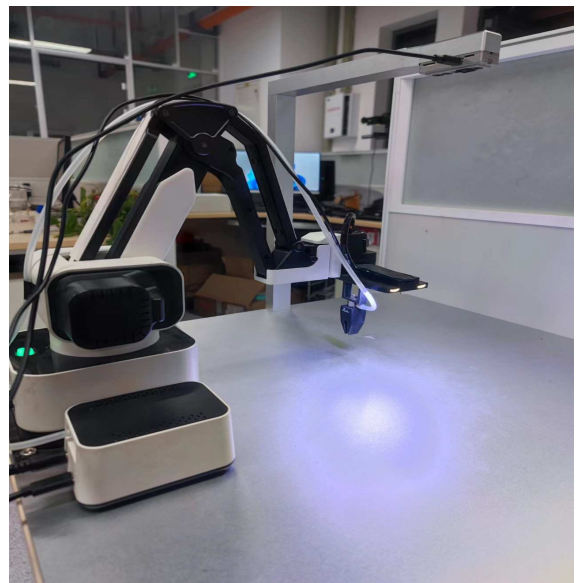


Figure 16. The Physical Appearance of RSSDR



Figure 17. Recognizing the Component



Figure 18. Gripping the Component

This paper proposes a robotic sorting system named RSSDR. Five modules of RSSDR, including a central host, robotics, vision components, microservices and machine vision control software. In the visualization toolkit, the 3D printed industrial parts dataset is constructed from 1760

images and evaluated with high accuracy and real-time using YOLOv5s.

Sorting robots based on machine vision technology can liberate workers from heavy labor and greatly improve the efficiency of sorting, so they are widely used in food, logistics, coal mining and other industries. The sorting of 3D printed industrial parts is only an application practice of this paper.

REFERENCES

- [1] Enzhang Jiao, Rong Du. Realization of industrial robot sorting rechnology[J].Modular Machine Tool & Automatic Manufacturing Technique,2000.
- [2] Song, Q.; Li, S.; Bai, Q.; Yang, J.; Zhang, X.; Li, Z.; Duan, Z. Object Detection Method for Grasping Robot Based on Improved YOLOv5. *Micromachines* 2021.
- [3] Kurihara, N., Nishikawa, M.;Watanabe;A.;Sato;Y.;Ohtsuka;K. A Combustion Diagnosis Method for Pulverized Coal Boilers using Flame-Image Recognition Technology[Z], 1986.
- [4] Xiang-qianwu, Kuan-quanwang, Davidzhang. Wavelet Energy Feature Extraction and Matching for Palmprint Recognition[Z], 2005.
- [5] Han SQ, Wang L. A Survey of Thresholding Methods for Image Segmentation[Z], 2002.
- [6] Baird, H.s. Model-based image matching using location.[Z], 1985.
- [7] Wan F, Minghui DU. Correction of Lens Distortion in Digital Camera Design[Z], 2005.
- [8] Jing Z, Xue FZ, Zu-shu LI. Space target location measurement based on monocular vision[Z], 2011.
- [9] Pahwa R. Depth camera calibration using depth measurements[Z], 2013.
- [10] Jian LI,Yang X,Bin HE. Research on RGB-D depth completion with deep learning[Z], 2018.
- [11] Redmon J, Farhadi A. YOLO9000: better, faster, stronger[C]// Proceedings of the IEEE conference on computer vision and pattern recognition. 2017.
- [12] Redmon J, Farhadi A. Yolov3: An incremental improvement[J]. arXiv preprint arXiv: 1804.02767, 2018.
- [13] BOCHKOVSKIY A, WANG C Y, LIAO H. YOLOv4:Optimal Speed and Accuracy of Object Detection[J].arXiv preprint arXiv: 2004.10934, 2020
- [14] Ling L,Tao J,Wu G. Research on Gesture Recognition Based on YOLOv5[Z].